X-Ray Calc 3 — User Manual

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1 Introduction

X-Ray Calc 3 (XRC3) is an open-source software package for simulating X-ray reflectivity (XRR) and solving the inverse problem of reconstructing film structures from measured XRR curves. It is designed for researchers working with X-ray optics, synchrotron radiation, and thin-film coatings.

The software employs Parratt's exact recursive method based on the Fresnel equations to calculate XRR and incorporates specialized tools for modeling periodic multilayer structures (PMMs). Refractive indices of materials are computed using the Henke tables of atomic scattering factors (Henke *et al.*, 1993). The computational algorithm has been validated against the *IMD* software (Windt, 1998), with accumulated variances consistently below 0.5%.

The application allows you to:

- Build multilayer structural models composed of stacks and individual layers
- Calculate theoretical X-ray reflectivity curves as a function of incidence angle (θ) or wavelength (λ)
- Load experimental reflectivity data and compare it with theoretical predictions
- Automatically fit model parameters to experimental data using the Shake LFPSO (Lévy Flight Particle Swarm Optimization) algorithm

- Visualize results through interactive charts with logarithmic/linear scaling
- Define parameter profiles (polynomial, exponential, tabulated) across periodic structures
- Manage materials via the built-in Henke atomic form factor database

> **Reference:** Penkov, O. V., Li, M., Mikki, S., Devizenko, A. & Kopylets, I. (2024). *X-Ray Calc 3: improved software for simulation and inverse problem solving for X-ray reflectivity.* J. Appl. Cryst. **57**, 555–566. https://doi.org/10.1107/S1600576724001031

2 Quick Start

This section provides a hands-on introduction to X-Ray Calc 3. The distribution includes several demonstration projects in the **Examples** folder — open one to explore the interface before building your own models.

Exploring the Demo Projects

1. Click **Open** and navigate to the **Examples** folder.
2. Select a project file (`.xrcx`).
3. Go to the **Computation** tab and click **Run** (or press **F5**). You can also use **Calc All** (**F12**) to calculate all models at once.

Navigating the Interface

- **Double-click a model name** in the Project Items list to open its Properties dialog (name, color, description). The same dialog is available from the right-click context menu.
- **Double-click a layer** to open the Layer Properties dialog, where you can change material, thickness, roughness, and density.
- **Double-click a stack header** to open the Stack Properties dialog, where you can rename the stack and change the number of periods.
- Use the **Stack** and **Layer** toolbar panels to add, insert, or delete structural elements.

Creating a Model from Scratch

1. Create a new project (**File > New Project**).
2. Select **Model 1** in the project tree.
3. Add a stack: **Structure > Add Stack** (or use the toolbar). In the dialog, set the stack name (e.g., “Main”) and the number of periods (default: 1 for non-periodic structures).
4. Select the stack by clicking its title in the structure panel.
5. Add layers: click **Add Layer** (or **Structure > Add Layer**). Fill in the material (click “...” to select from the Materials Database), thickness, roughness, and density. Repeat for each layer.
6. Calculate the reflectivity: press **F5** (or **Calc > Calculate**).

Calculation Settings

- **Mode:** Reflectivity can be calculated as a function of grazing angle or wavelength.
- **Polarization:** Use **s-polarization** for hard X-rays at low grazing angles (below 20°) — this is significantly faster. Use **sp-polarization** for larger angles or softer radiation.
- **Number of points (N):** Controls calculation precision. Calculation time scales linearly with N.
- **Angle range:** Set with θ_1 and θ_2. Check **2θ** for θ-2θ geometry. **dθ** sets the instrumental beam divergence.
- **Scale:** Toggle between **Linear** and **Log** scale using the toolbar button. Set a **background level** if needed.
- **Legend:** Click items in the chart legend to toggle curve visibility.

Exporting Results

- The **Export** button on the Result panel saves the calculated curve to an ASCII file or copies it to the clipboard.
- The **Save** button on the Plot panel saves the chart as a bitmap image.

3 Installation & First Launch

System Requirements

- Windows 10 or later (64-bit recommended)
- Minimum 4 GB RAM (8 GB recommended for large models)
- Multi-core CPU recommended for parallel computation

Data Storage

On first launch, X-Ray Calc 3 creates a working directory:

- **Default location:** `%APPDATA%\X-RayCalc3\`
- **Portable mode:** If a file named `uselocaldata` exists in the application folder, all data is stored locally next to the executable.

The working directory contains:

File / Folder	Purpose
`xrc3.ini`	Application settings
`Henke/`	Atomic form factor tables
`Backup/`	Auto-save and backup projects
`Output/`	Fitting results (when auto-save is enabled)
`Benchmark/`	Benchmark model files
`Jobs/`	Batch job definitions

File Association

To associate `.xrcx` project files with X-Ray Calc 3, go to **File** > **Settings** > **Paths and Files** and click **Register file associations (xrcx)**.

4 Application Overview

Main Window Layout

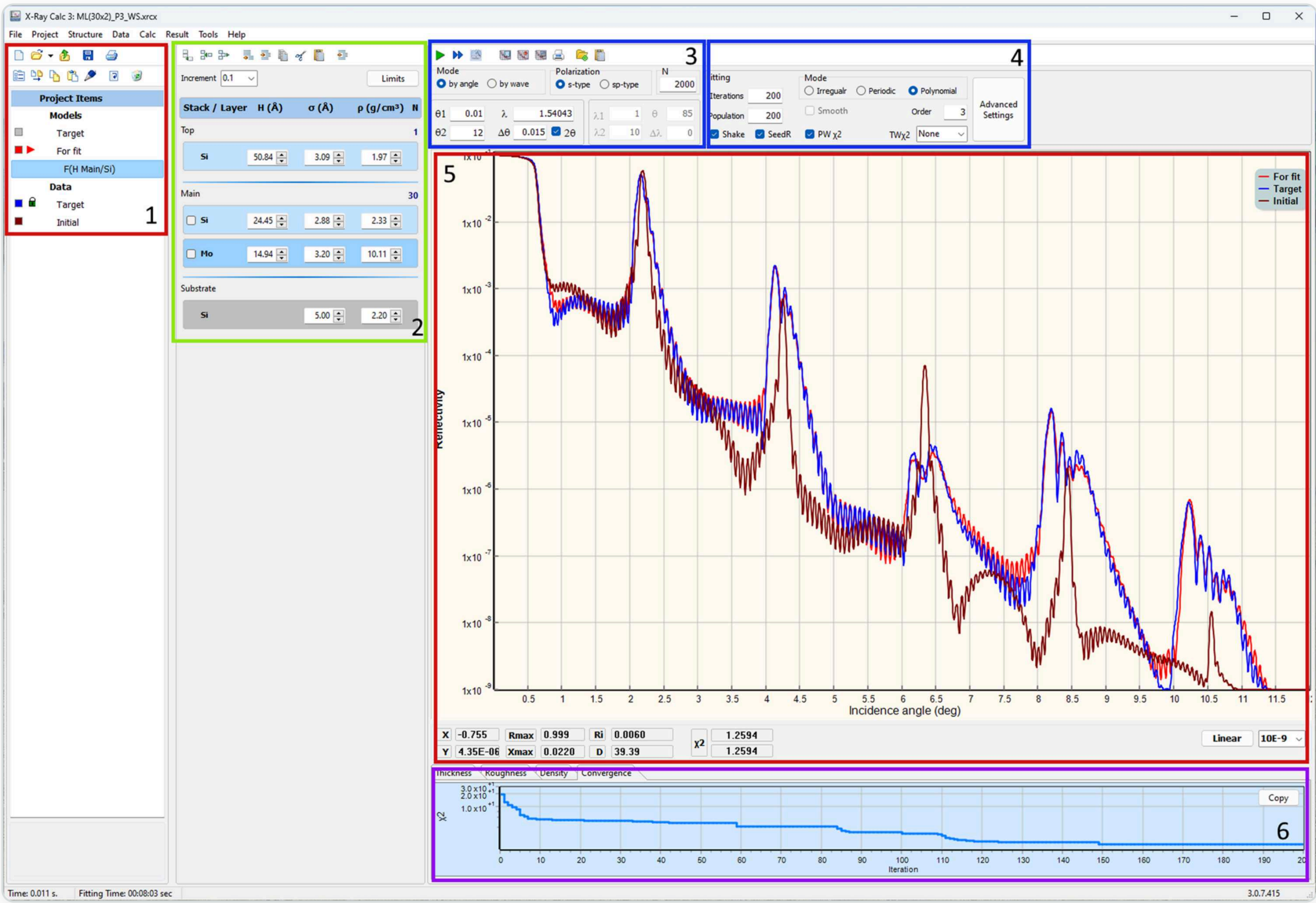

Figure 1. *Screenshot of the XRC3 GUI. (1) Project tree. (2) Structure panel. (3) XRR calculation parameters. (4) Fitting parameters. (5) XRR curves. (6) Distributions / Convergence panel. (From Penkov et al., 2024.)*

The primary window comprises four key panels (see Figure 1). The first panel, **Project Items** (1), showcases the contents of the current project. A project can organize an unlimited number of theoretical models and experimental curves into well-structured folders. The **Structure panel** (2) functions as a visual depiction of the layered model and provides easy access to all parameters linked with the structural model. The right side contains **XRR calculation parameters** (3) and **Fitting parameters** (4) at the top, the main **XRR curve plot** (5) in the center, and **Distributions / Convergence charts** (6) at the bottom.

Left Column — Project Panel

- **File Toolbar:** New, Open, Reopen, Save, Print

- **Project Toolbar:** Add Model, Export, Copy, Paste, Edit Properties, Add Extension, Copy Image, Print, Delete
- **Project Tree:** Hierarchical view of all models, data sets, and folders in the current project
- **Description:** Read-only text memo showing the description of the selected item

Middle Column — Structure Panel

- **Structure Toolbar:** Stack operations (Add, Insert, Delete) and Layer operations (Add, Insert, Copy, Cut, Paste, Delete)
- **Increment Selector:** Step size for quick parameter adjustment (10, 5, 1, 0.25, 0.1, 0.01)
- **Set Fit Limits Button:** Opens the parameter bounds editor for fitting
- **Visual Structure Editor:** Interactive graphical representation of the layer structure — shows Ambient, Stacks (with N repetitions), individual Layers (with H, σ, ρ values), and Substrate. Double-click elements to edit properties; click parameter values for in-place editing.

Right Column — Working Area

- **Calculation Toolbar:** Load Data, Paste Data, Calculate, Save Result, Copy Result, Batch Execute, Auto Fit, Stop
- **Settings Panel:** Calculation mode and parameters (left half) alongside Fitting mode and parameters (right half)
- **Main Chart:** Interactive reflectivity plot (θ or λ vs. reflectivity)
- **Chart Info Bar:** Cursor position, peak detection, area integration, χ^2 values
- **Chart Pages:** Tabbed sub-charts for parameter distributions (Thickness, Roughness, Density, Profile, Fitting Progress)

Status Bar

Displays calculation time, fitting time, application version, and platform (x64).

Menu Bar

Menu	Purpose
File	Project management (New, Open, Save, Settings, Exit)
Project	Model operations (New Model, Duplicate, Copy, Paste, Add Folder)
Structure	Stack and layer editing (Add, Insert, Delete, Copy, Paste, Cut)
Data	Experimental data operations (Load, Paste, Normalize, Smooth, Trim, Export)
Calc	Calculation and fitting (Calculate, Calculate All, Auto Fitting, Batch Jobs, Benchmark)
Result	Export results (Save, Copy to Clipboard, Copy as BMP/WMF)
Tools	Material management (Create Material, Edit Henke Table)
Help	User Manual, About dialog

5 Projects

Project Files

Projects are saved as `.xrcx` files. Internally, an `.xrcx` file is a ZIP archive containing:

- `project.dsc` — project metadata
- `params.dsc` — model structure definitions
- Associated experimental data files

Creating a New Project

1. Choose **File > New Project** (or click the **New** toolbar button).
2. An empty project is created with no models or data.

Opening a Project

- **File > Open Project** — browse for a `.xrcx` file
- **File > Recent Projects** — select from up to 10 recently opened projects
- **Open toolbar button** — click the dropdown arrow for recent projects

Saving a Project

- **File > Save Project** — save to the current file
- **File > Save Project As** — save with a new name/location

Auto-Save

When enabled in Settings, the application periodically saves a backup to `AutoSave.xrcx` in the temp directory. This provides crash recovery — on next launch, you will be prompted to restore the auto-saved project.

Project Locking

When a project is open, a lock file (`temp.lock`) prevents simultaneous access from another instance. The lock is released when the project is closed.

Project Tree Organization

The project tree shows items in a hierarchy:

- **Model Group:** Contains all structural models
  - Individual models (each with its own layer structure and calculation parameters)
  - Extensions (profile functions attached to models)
- **Data Group:** Contains experimental data sets
- **Folders:** User-created organizational folders

> **Tip:** Right-click the project tree for context menu operations.

6 Defining a Model

Creating a Model

1. Choose **Project > New Model** or click the **Add Model** toolbar button.
2. A new model appears in the project tree with a default empty structure.

Duplicating a Model

Select a model in the project tree and choose **Project > Duplicate Model** to create an exact copy, including all layers, stacks, and parameters.

The Structure Editor

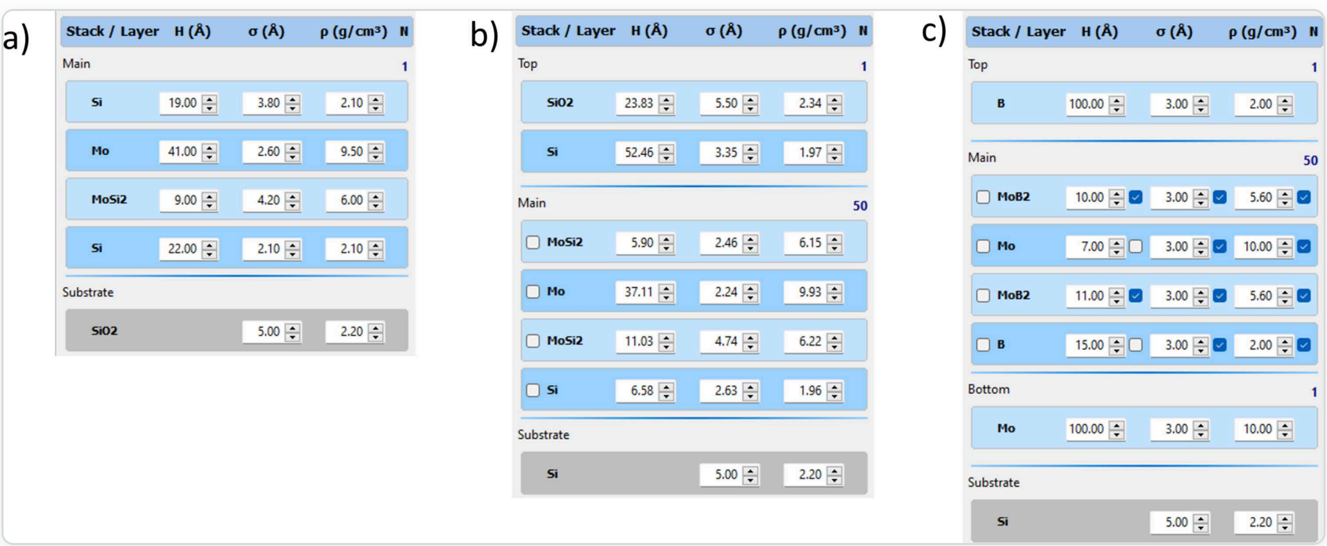

Figure 2. *Examples of different structure types. (a) Regular structure. (b) Periodic structure. (c) Periodic structure with distributions. (From Penkov et al., 2024.)*

The structure editor is a visual panel in the middle column that displays the current model's layer structure. It shows:

- **Ambient** (top) — the medium above the structure (typically vacuum or air)
- **Stacks** — repeating periodic units containing one or more layers
- **Individual Layers** — single layers outside of stacks
- **Substrate** (bottom) — the base material

Each element in the structure is displayed as a colored block with its parameters.

Terminology

- **Material** — the chemical composition of a layer (e.g. Mo, SiO_2, B_4C). Used to look up atomic scattering factors, atomic weights and table densities in the embedded database.
- **Layer** — the basic geometrical entity. A layer consists of a material and has three parameters: thickness (H), roughness (σ) and density (ρ).
- **Stack** — a set of layers. It can be single (non-repeating) or periodic (repeated N times).
- **Structure** — a set of stacks, including a substrate. The substrate is a special stack consisting of a single layer with infinite thickness.

Stacks (Periods)

A stack represents a periodic multilayer structure — a group of layers repeated N times.

Adding a stack

1. Select the model in the project tree.
2. Choose **Structure > Add Stack** or use the structure toolbar.
3. The Stack Properties dialog opens with:
   - **Stack name:** A label for the stack
   - **N:** Number of repetitions (periods)

Editing a stack

Double-click the stack header in the structure editor to open the Stack Properties dialog.

Layers

Each layer has three primary parameters:

Parameter	Symbol	Unit	Description
Thickness	H	Å (Angstrom)	Physical thickness of the layer
Roughness	σ	Å (Angstrom)	Interface roughness (RMS)
Density	ρ	g/cm^3	Material density

Adding a layer

1. Select a stack or model in the structure editor.
2. Choose **Structure > Add Layer** or use the toolbar.
3. The Layer Properties dialog opens with fields for material, H, σ, and ρ.

Layer operations (via Structure menu or toolbar)

- **Insert Layer** — insert before the selected position
- **Copy Layer** — copy to internal clipboard
- **Paste Layer** — paste from internal clipboard
- **Cut Layer** — copy and delete
- **Delete Layer** — remove the layer

Quick Parameter Editing

In the structure editor, you can adjust layer parameters directly:

- Click a parameter value to edit it in-place
- Use the **Increment** selector to set the step size for fine-tuning
- Changes are applied immediately; if Auto-Calc is enabled, the reflectivity recalculates in real time

Undo

Choose **Structure > Undo Structure Changes** to revert the last structural modification.

Model Text (JSON)

For advanced users, **Structure > Edit Model Text** opens a JSON editor showing the raw model definition.

7 Experimental Data

Loading Data from File

1. Select a model or data item in the project tree.
2. Choose **Data > Load from File** or click the **Load Data** toolbar button.
3. Browse for a text file containing two-column data (angle/wavelength and reflectivity).
4. The data appears as a series on the main chart.

Pasting Data from Clipboard

1. Copy two-column data from another application (e.g., a spreadsheet).
2. Choose **Data > Paste from Clipboard** or click the **Paste Data** toolbar button.
3. The data is imported into the current model.

Data Format

The expected data format is a two-column ASCII text file:

- **2 columns** with TAB as the column separator
- **3-row header** (non-numeric lines are skipped automatically)
- Column 1: Angle (degrees) or wavelength (Angstroms)
- Column 2: Reflectivity (typically 0 to 1)

Example:

```
Sample: Mo/Si multilayer
Date: 2024-01-15
Theta    Reflectivity
0.100   1.000000
0.200   0.999850
0.300   0.998200
0.500   0.956300
1.000   0.523100
2.000   0.012450
5.000   0.000023
```

> **Tip:** You can also copy and paste table data directly from **OriginLab Origin**.

Data Processing

Normalize

Data > Normalize opens a dialog where you can manually set the normalization factor. **Data > Normalize (Auto)** automatically normalizes the data.

Smooth

Data > Smooth applies a Savitzky-Golay smoothing filter to reduce noise while preserving peak shapes.

Trim

Data > Trim removes the low-reflectivity tail of the data.

Data Conditioning Walkthrough

Raw experimental data often requires conditioning before fitting:

Step 1 — Normalize. Compare experimental and calculated intensities at a known point. Apply via **Data > Normalize**.

Step 2 — Smooth. Apply smoothing to reduce noise: **Data > Smooth**.

Step 3 — Trim. Remove data points outside the useful range: **Data > Trim**.

Exporting Data

- **Data > Copy to Clipboard** — copy data in tab-separated format
- **Data > Export to File** — save processed data to a text file

8 Calculation

Algorithm

XRC3 uses the Parratt exact recursive method for simulating specular X-ray reflectivity. A Gaussian distribution is assumed for interface roughness. The refractive indices of materials are calculated using the Henke tables of X-ray scattering factors.

Calculation Modes

Theta Mode (by angle)

Calculates reflectivity as a function of incidence angle at a fixed wavelength.

Parameter	Description
θ_1	Start angle (degrees)
θ_2	End angle (degrees)
λ (Å)	Fixed wavelength in Angstroms (default: 1.54043 Å, Cu Kα)
Δθ	Angular resolution / step size (degrees)
2θ	When checked, angles are in 2θ convention

Lambda Mode (by wavelength)

Calculates reflectivity as a function of wavelength at a fixed incidence angle.

Parameter	Description
λ_1	Start wavelength (Angstroms)
λ_2	End wavelength (Angstroms)
θ	Fixed incidence angle (degrees)
Δλ	Wavelength step size (Angstroms)

Polarization

- **s-type:** S-polarization (TE mode)
- **sp-type:** Mixed σ/π polarization

> **Guidance:** Use **s-polarization** for hard X-rays at low grazing angles (below 20°) — significantly faster. Use **sp-polarization** for larger angles and softer radiation.

Running a Calculation

1. Select a model in the project tree.
2. Set the calculation parameters.
3. Choose **Calc > Calculate** or click the **Calc Run** toolbar button.
4. The calculated reflectivity curve appears on the main chart.

Multi-Threading

The angular range is divided into segments matching CPU cores for parallel computation. Configure in **File > Settings > Calc & Fit**.

9 Fitting (Optimization)

Fitting adjusts model parameters to minimize the difference between calculated and experimental reflectivity curves. X-Ray Calc 3 uses the **Shake LFPSO (Lévy Flight Particle Swarm Optimization)** algorithm.

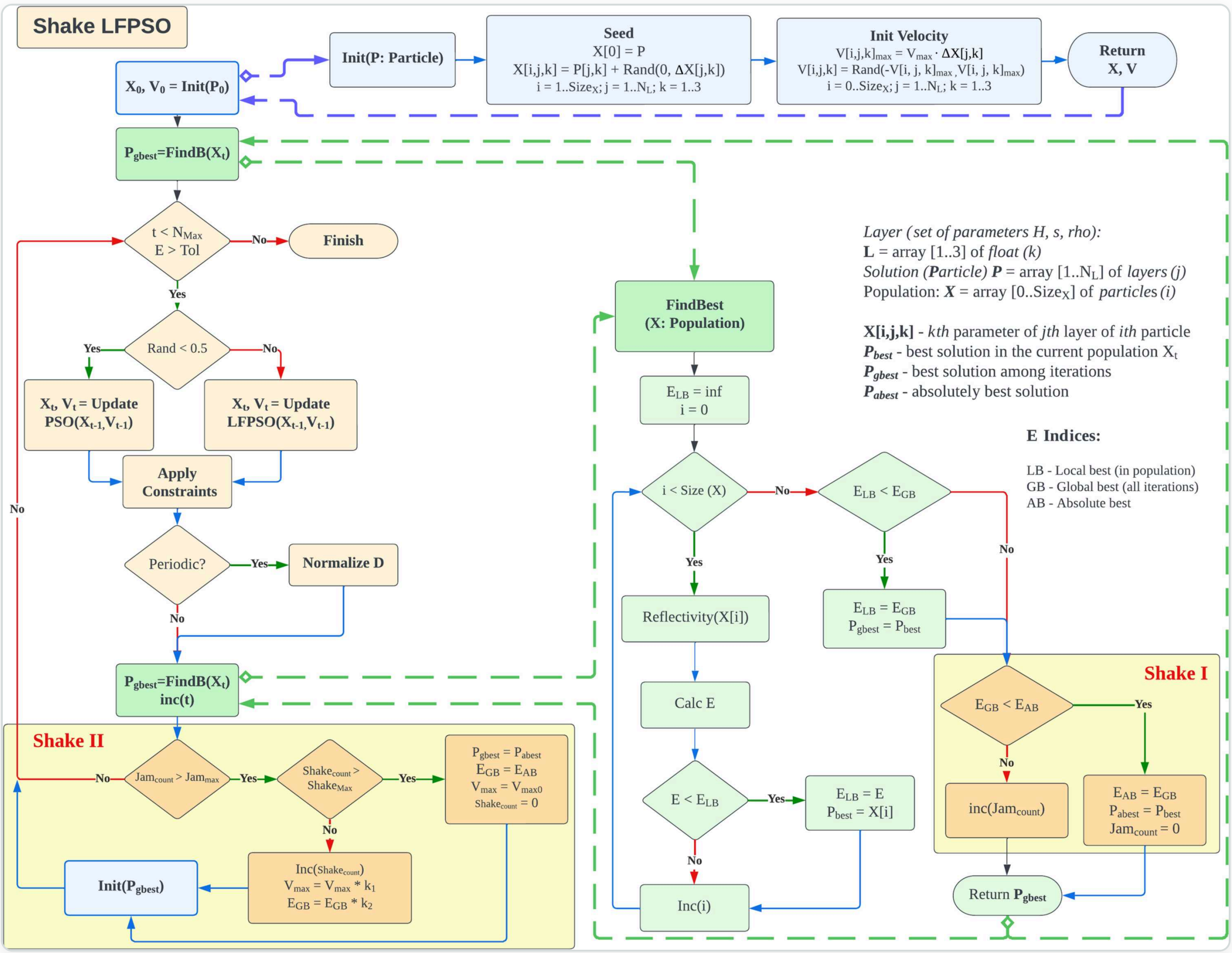

Figure 3. *General diagram of the Shake LFPSO algorithm. (From Penkov et al., 2024.)*

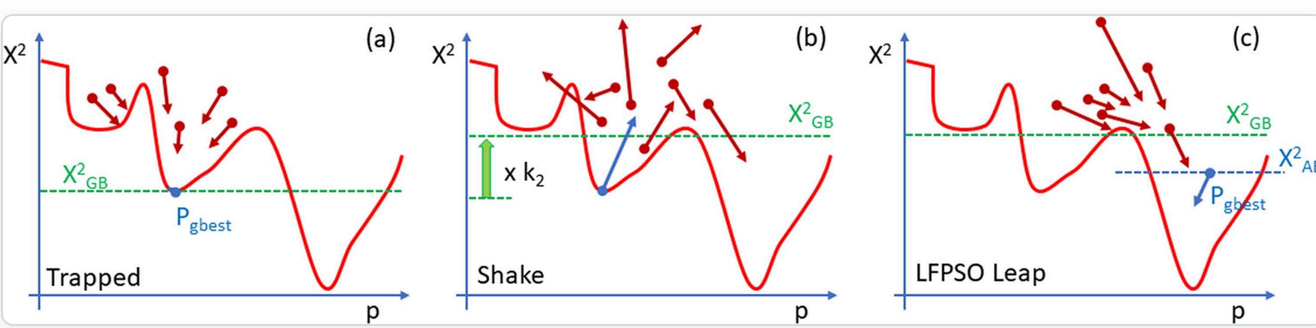

Figure 4. *The Shake concept. (From Penkov et al., 2024.)*

Prerequisites for Fitting

1. **A defined model** with initial parameter estimates
2. **Experimental data** loaded into the project
3. **Fit limits** set for each adjustable parameter

Setting Fit Limits

Click **Set Fit Limits** in the structure toolbar. Only parameters with Min ≠ Max will be adjusted during fitting.

Fitting Modes

Structure Representation	Applicability	Computation Time
Regular (Irregular)	Non-periodic coatings	Long
Periodic	Periodic multilayer mirrors	Short
Periodic with arbitrary deviations	Supermirrors	Long
Polynomial distributions	PMMs with gradual parameter change	Moderate

Basic Fitting Parameters

Parameter	Description	Default
Iterations	Maximum optimization iterations	100
Population	Number of particles in the swarm	100
Shake	Enable perturbation to escape local minima	Checked
SeedR	Seed particles from parameter range	Checked
Smooth	Apply smoothing to fitted profiles	Unchecked
PW χ^2	Point-weighted chi-square	Checked
TW χ^2	Theta-weighted chi-square	None

Advanced Fitting Settings

Parameter	Description	Default
Tolerance	Target cost function value	0.005
SHmax	Maximum consecutive shakes	3
Jmax	Iterations without improvement before shake	1
Vmax	Maximum particle velocity factor	0.3
ω_1	Velocity scale factor	0.3
ω_2	Velocity reducing factor	0.1
k_1	Velocity shake coefficient	1.41
k_2	Cost function shake coefficient	1.41

Running a Fit

1. Ensure a model with experimental data and fit limits is selected.
2. Configure fitting mode and parameters.
3. Choose **Calc > Auto Fitting** or click the **Fast Forward** toolbar button.
4. Fitting stops when iterations are exhausted, tolerance is met, or you click **Stop**.

Understanding the Cost Function

$$E = \Sigma \left|\log I(\theta)_{exp} - \log I(\theta)_{calc}\right| \times \omega(\theta)_{peak} \times \omega(\theta)_{\theta}$$

Lower is better — E = 0 means a perfect fit. Calculated on a logarithmic scale to give equal weight to both high and low reflectivity regions.

10 Results & Export

Saving Results

Result > Save Results saves the calculated reflectivity data to a text file.

Copying to Clipboard

- **Result > Copy to Clipboard** — tab-separated text
- **Result > Copy as BMP** — bitmap image
- **Result > Copy as WMF** — Windows Metafile (vector format)

Auto-Save Results

When enabled in Settings, fitting results are automatically saved to the configured Output directory after each fitting run.

11 Charts & Visualization

Main Reflectivity Chart

- **Zoom:** Click and drag to select a zoom region
- **Pan:** Right-click and drag
- **Reset zoom:** Double-click
- **Scale toggle:** Switch between linear and logarithmic Y-axis

Chart Pages

Below the main chart are tabbed sub-charts: **Thickness (H)**, **Roughness (σ)**, **Density (ρ)**, **Profile**, and **Fitting Progress**.

Peak Analysis

- **Peak Max:** Maximum reflectivity value and position
- **Area:** Integrated area under the peak
- **Period:** Estimated multilayer period from peak spacing

12 Profile Extensions

Profile extensions define how layer parameters vary across a periodic structure.

Profile Types

Function	Formula	Use Case
Polynomial	$a_0 + a_1x + a_2x^2 + \ldots$	General-purpose gradient
Exponential	$a_0 \cdot e^{a_1x}$	Exponential growth/decay
Parabolic	$a_0 + a_1x^2$	Symmetric variation
SQRT	$a_0 + a_1\sqrt{x}$	Gradual initial change
Table	(position, value) pairs	Arbitrary profile shapes

13 Tools

Create New Material

Tools > Create New Material adds a custom material to the Henke database by computing atomic form factors from constituent elements.

Edit Henke Table

Tools > Edit Henke Table opens the Henke Table Editor for viewing and modifying atomic scattering factor tables.

Benchmark

Calc > Benchmark measures calculation performance by running a standard reflectivity calculation multiple times.

14 Application Settings

Access via **File > Settings**. Sections: Paths and Files, Behavior, Interface, Calc & Fit, Graphics.

Behavior

Setting	Default
Automatically calculate when opening project	On
Live update structure panel during fitting	Off
Automatically save results to output folder	Off
Automatically check for updates	Off

15 Quick Reference

Toolbar Quick Reference

Button	Action
New	Create a new project
Open	Open an existing project
Save	Save the current project
Add Model	Add a new model
Load Data	Load experimental data
Calc Run	Calculate the selected model
Fast Forward	Start auto-fitting
Stop	Abort calculation or fitting
Scale	Toggle linear/log chart scale

Keyboard Shortcuts

Key	Action
F1	Open User Manual

16 File Formats

Project File (.xrcx)

A ZIP archive containing `project.dsc`, `params.dsc`, and embedded experimental data files.

Experimental Data Files

Plain text, two columns separated by tabs or spaces. Lines starting with `#` are comments.

Henke Tables

Binary files in the `Henke/` directory containing wavelength-dependent atomic form factors (f_1, f_2).

Settings File (xrc3.ini)

Standard Windows INI file. Sections: `[Path]`, `[Window]`, `[Options]`, `[Graphics]`, `[Calc]`.

17 Illustrative Examples

The following examples from the literature demonstrate XRC3 capabilities (Penkov *et al.*, 2024).

Single-Layer Coating

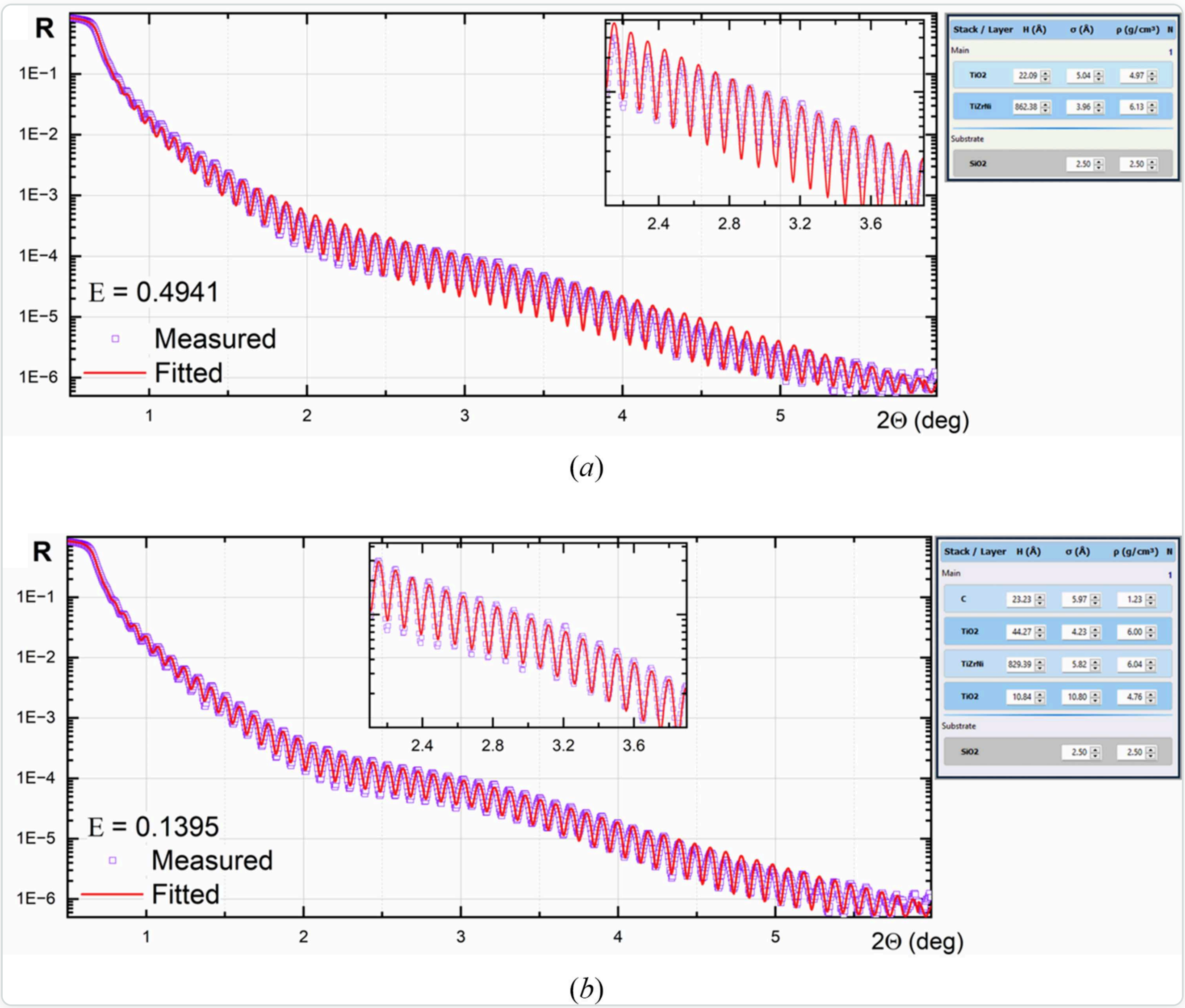

Figure 5. *XRR curve fitting for a single-layer coating. (From Penkov et al., 2024.)*

Periodic X-Ray Mirrors

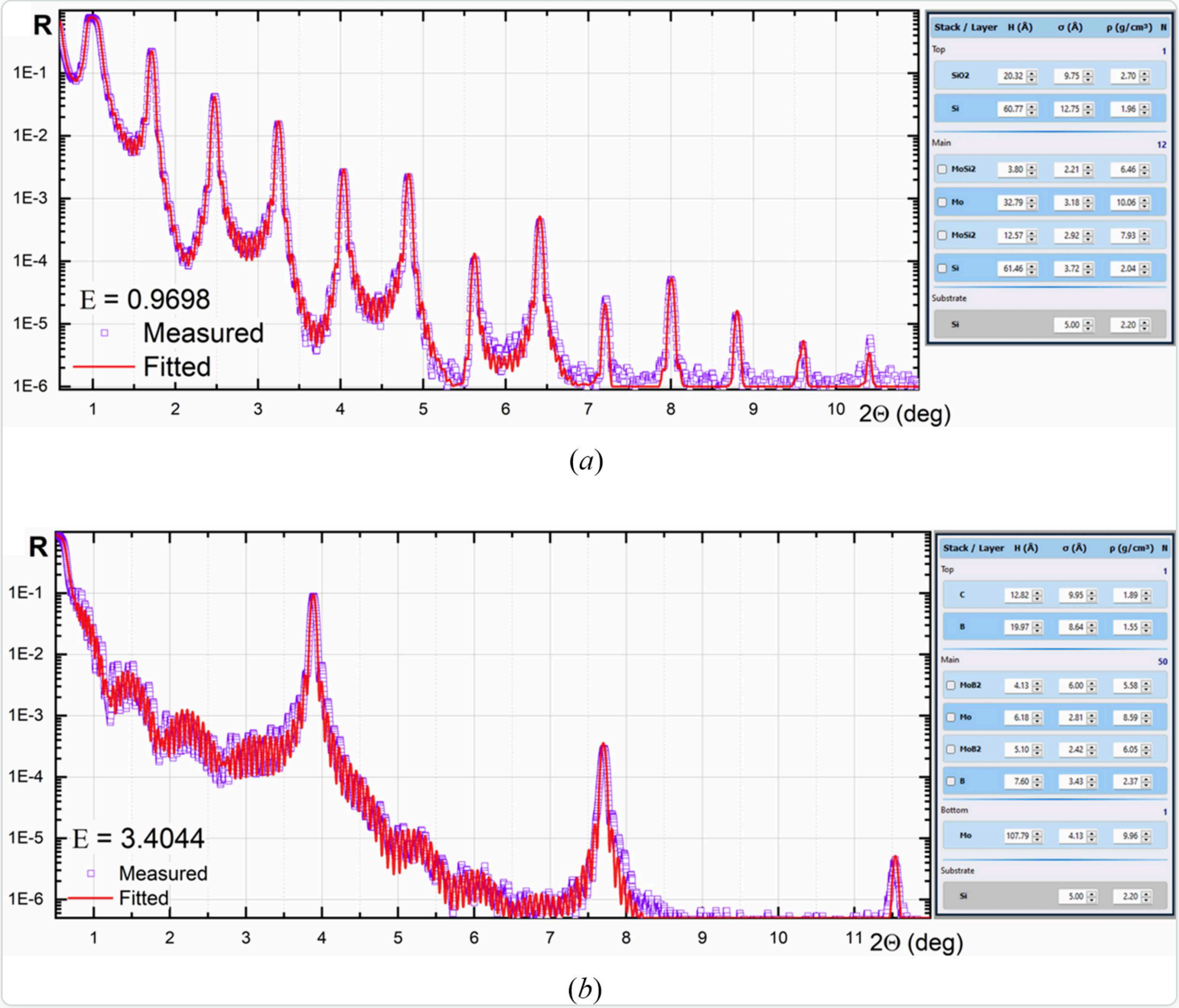

Figure 6. *XRR curve fitting for X-ray mirrors. (a) Mo/Si. (b) Mo/B. (From Penkov et al., 2024.)*

Polynomial Fitting on Graded PMMs

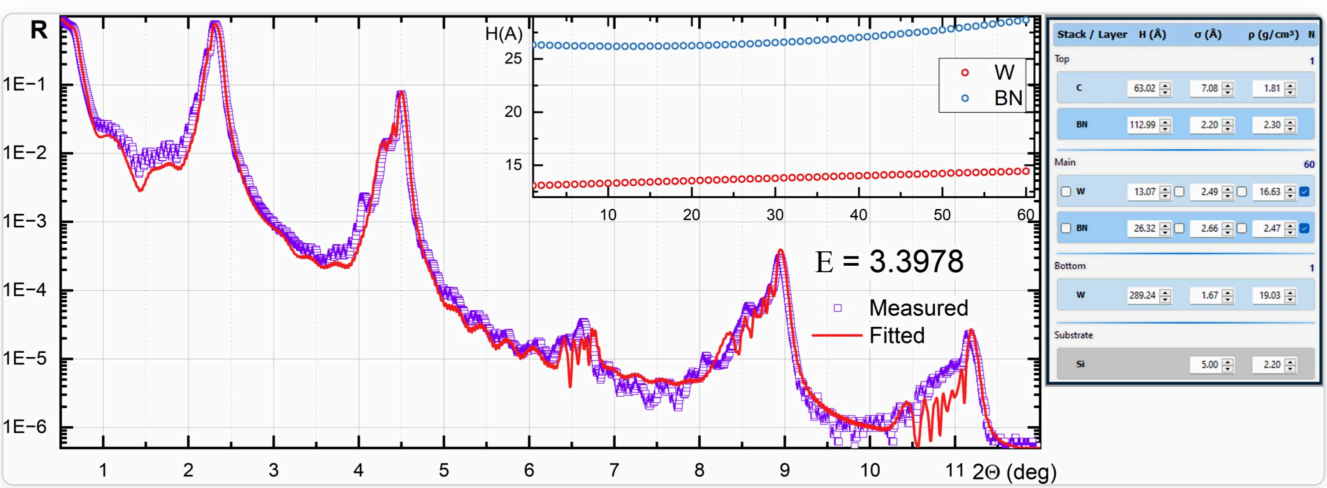

Figure 7. *Polynomial XRR curve fitting for a W/BN X-ray mirror. (From Penkov et al., 2024.)*

Supermirror

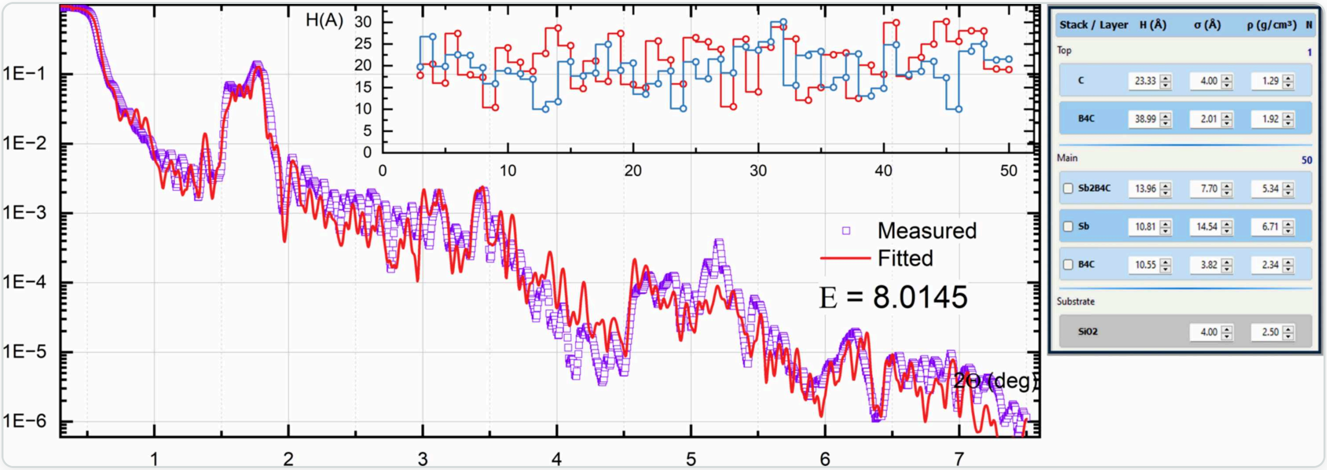

Figure 8. *XRR curve fitting for an Sb/B_4C supermirror. (From Penkov et al., 2024.)*

18 Troubleshooting

Application crashes without error message

Solution: Use the **x64 version** (`XRayCalc3.x64.exe`). The x32 version is limited to ~2 GB RAM.

Choosing between x32 and x64

Version	Advantage	Limitation
x32	~2× faster calculation	Limited to ~2 GB RAM
x64	Handles large models	Slower for small models

Fitting does not converge

- Widen the fit limits
- Increase population size
- Increase iterations
- Enable Shake
- Check initial model
- Try a different fitting mode

Calculation is slow

- Ensure multi-threading is enabled
- Reduce number of points (N)
- Use x32 for small models (~2× faster)
- Use x64 only for large models

Material not found

Check spelling. Use **Tools > Create New Material** to add custom compounds.

Chart appears empty after calculation

Check angle/wavelength range. Try switching between linear and logarithmic scale. Verify the model has at least one layer.

Data import fails

Ensure plain text with two numeric columns. Check decimal separators match your locale. Try clipboard paste instead.

X-Ray Calc 3 — Oleksiy Penkov — oleksiypenkov@intl.zju.edu.cn

Penkov, O. V., Li, M., Mikki, S., Devizenko, A. & Kopylets, I. (2024). *J. Appl. Cryst.* **57**, 555–566. doi:10.1107/S1600576724001031